# Junkyard Prison — Game Design Document

Oct 23, 2026 · @seki

Game Design Document for Junkyard Prison, a single-player Roblox simulator. Version 1.1.


| Part I — Summary | Part II — Detailed design |
|---|---|
| 1. Game overview | 12. Feature specifications |
| 2. Story and setting | 13. Economy and balancing |
| 3. Core gameplay loop | 14. UI screen specifications |
| 4. Trash and sorting | 15. Narrative content |
| 5. Progression | 16. Audio list |
| 6. Economy and upgrades | 17. Technical appendix |
| 7. Monetization | 18. Analytics and KPIs |
| 8. World and level design | 19. Accessibility, localization and platforms |
| 9. Onboarding, UI and audio | 20. QA test plan |
| 10. Technical design | 21. Roadmap, glossary and version history |
| 11. Launch checklist and risks | |


## 1. Game overview

Junkyard Prison is a single-player Roblox simulator. The player is jailed in a wire-cage landfill and walks free only after hand-clearing exactly 100,000 pieces of trash into the right bins.



| Field | Value |
|---|---|
| Title | Junkyard Prison (working title was Trash Prison Simulator) |
| Platform | Roblox: PC, mobile, tablet, console |
| Genre | Simulator / idle-progression, light puzzle (sorting) |
| Players | 1 per server (Maximum Visitor Count = 1) |
| Audience | Ages 9+ target, launching first to the 16+ verified audience |
| Session length | 10–30 min sessions over a multi-week grind |
| Win condition | Sort all 100,000 items and walk out the gate |
| Developer | Seki (solo) |

## Elevator pitch

You littered once too often. Now the Warden has locked you in a landfill with a mountain of 100,000 pieces of trash. Pick it up, sort it into Recyclable, Biodegradable or Non-Biodegradable, and earn coins for better tools until the pile is gone and the gate opens.

## Design pillars

1. **A real, finite pile.** Every one of the 100,000 items is a physical object in the world. The mountain visibly shrinks, and the count is always exact.
2. **Sorting is the skill.** The right bin pays and counts toward freedom; the wrong bin is rejected. Players learn real-world waste categories by doing.
3. **Steady, visible power growth.** Bare hands, then a sack, then tools that grab more pieces faster. Each upgrade changes how the pile feels to clear.
4. **Fair, optional monetization.** Everything can be earned in play. Robux purchases save time but never gate the ending.

## Unique selling points

- A single 100k-piece pile that persists between sessions per save slot.
- Educational sorting wrapped in a prison-escape story.
- A clear end state, which is rare for Roblox simulators.

## Store page assets

The game icon (512 × 512) shows the player in a prison jumpsuit in front of the pile. Five in-game thumbnails (1920 × 1080) appear through this document: the pile (sections 1 and 8), the sorting station (section 4) and the night shift (section 8).




## 2. Story and setting

The player is sentenced to Junkyard Prison for repeated littering and for ignoring every warning and fine. The sentence is simple: clear 100,000 pieces of trash, and the gate opens.




## The Warden

The Warden is the game's only speaking character. He is never seen, only heard over the prison PA system. He welcomes the inmate, reads the sentence and coaches the first trip, then goes quiet. His tone is stern but fair, never cruel.

## Setting

- **The yard:** a long wire-mesh cage built over a desert landfill, open to the sky.
- **The pile:** one mountain of everyday trash filling most of the cage, up to about 74 studs high.
- **The sorting station:** three color-coded bins beside the gate, under a "Which bin?" guide board.
- **Outside:** a desert road, power lines and empty sand, visible through the mesh as a reminder of freedom.

## Tone

Light, colorful and a little cheeky. The "prison" is a frame for the goal, not a source of threat. There is no violence, no guards to fight and no punishment beyond a rejected item.

### Ending

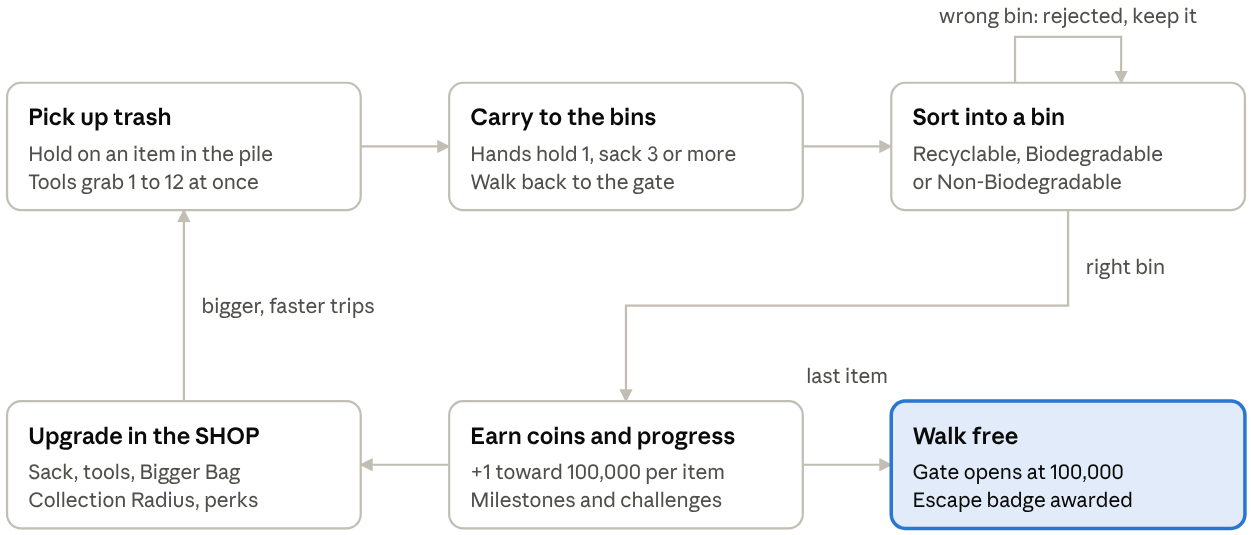
When the 100,000th item is sorted, the inmate has served the sentence. A YOU ESCAPED! screen offers WALK OUT (guided to the open gate) or KEEP PLAYING, and the escape badge is awarded once its ID is set.

After escaping, an optional rebirth resets the run for a permanent +25% coin bonus per rebirth.

## 3. Core gameplay loop

The loop is collect, carry, sort, earn, upgrade. Each trip gets bigger and faster as tools and the sack improve.

**Each sorted item pays coins that buy faster trips**




core loop · 5 steps and the exit

A wrong bin costs only the walk; every right bin moves the counter toward 100,000 and funds the next upgrade.

### Player actions

| Action | PC | Mobile / console | Notes |
|---|---|---|---|
| Move, jump, camera | WASD, Space, mouse | Thumbstick, jump button, drag | Standard Roblox controls |

| Action | PC | Mobile / console | Notes |
|---|---|---|---|
| Pick up trash | Hold E on an item | Hold the on-screen prompt | Hold time = the tool's pickup time (0.6 s base, 0.12 s minimum) |
| Deposit | Hold E at a bin | Hold the bin prompt | All matching items in the sack go in at once |
| Open menus | SHOP, GOALS, DAILY, MENU buttons | Same buttons | Right-side HUD stack |
| Save / exit | SAVE, EXIT buttons | Same | Exit returns to the main menu |

### Carrying

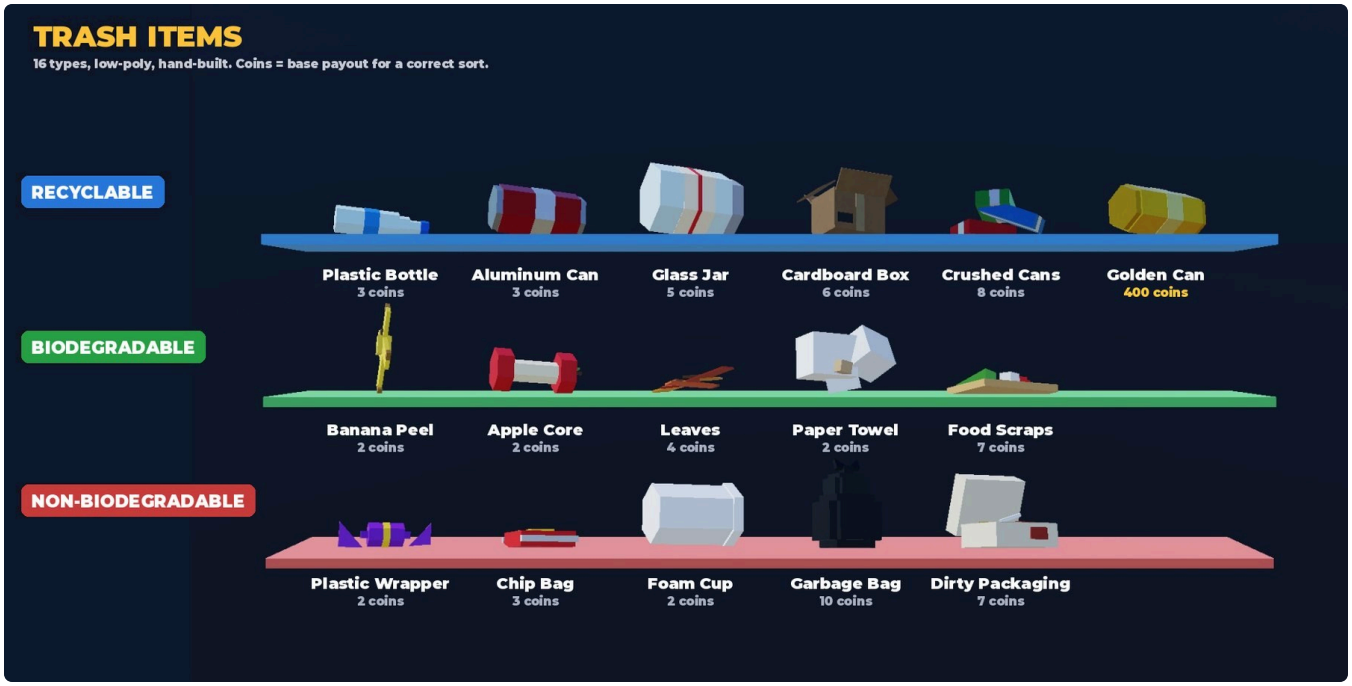
- Bare hands carry 1 item.
- The **Trash Sack** carries 3, and the **Bigger Bag** upgrade raises it further.
- Tools grab several nearby items in one pickup, up to the free space in the sack.

## 4. Trash and sorting

Every new game starts with exactly 100,000 physical trash items in one pile, and the game guarantees the count never drifts: pile + loose + carried + sorted always equals 100,000.

### Categories and bins

| Bin | Color | Items |
|---|---|---|
| Recyclable | Blue | Plastic Bottle, Aluminum Can, Glass Jar, Cardboard Box, Crushed Cans, Golden Can (rare) |
| Biodegradable | Green | Banana Peel, Apple Core, Leaves, Paper Towel, Food Scraps |
| Non-Biodegradable | Red | Plastic Wrapper, Chip Bag, Foam Cup, Garbage Bag, Dirty Packaging |




Base payouts run from 2 coins (peels, wrappers, foam cups) to 10 (garbage bags); the rare Golden Can pays 400. Coin Multiplier, Sorting Efficiency, passes and boosts multiply these.

## Sorting rules

- **Right bin:** the item counts toward the 100,000 and pays coins.
- **Wrong bin:** the item is rejected and stays with the player. Nothing is lost.
- **Golden cans:** rare items hidden in the pile that are worth a large coin payout.




### The pile

- Items are low-poly, hand-built models (about 68–212 triangles each) so all 100,000 can be drawn at once.
- The pile is split into 40-stud chunks. Items stay anchored until the player disturbs them, then fall with physics.
- Items knocked loose stay where they land and are saved as loose items.
- Each save slot keeps its own pile, so a returning player finds the mountain exactly as they left it.

## 5. Progression

Progress is measured by one number, items sorted out of 100,000, shown on the HUD and on the progress board above the gate. Milestones, challenges and daily rewards hang off that number.

### Key unlock points

| Items sorted | What happens |
|---|---|
| 25 | First milestone |

| Items sorted | What happens |
|---|---|
| 50 | Trash Sack can be bought (SHOP > TOOLS) |
| 100 | First Shift milestone, rewards the Basic Grabber (usable once the sack is owned) |
| 25,000 | Advanced Waste Extractor can be bought |
| 35,000 | Auto-Deposit perk can be bought |
| 50,000 | Auto-Sort Sack perk can be bought |
| 100,000 | Sentence served: gate opens, escape badge |

## Milestones

18 milestones, each shown as a banner with a sound sting. Rewards grow from pocket change to six figures, with titles, trails and permanent coin bonuses at key points.

| Items sorted | Milestone | Reward |
|---|---|---|
| 25 | Fresh Inmate | 25 coins |
| 100 | First Shift | Basic Grabber + 100 coins |
| 250 | Getting the Hang | 250 coins |
| 500 | Trash Rookie | 500 coins |
| 1,000 | Yard Sweeper | 1,000 coins |
| 2,500 | Bin Regular | 2,000 coins |
| 5,000 | Landfill Veteran | 4,000 coins |
| 7,500 | Dumpster Diver | 6,000 coins |
| 10,000 | Clean Freak | "Clean Freak" title + Sparkle trail |
| 15,000 | Model Prisoner | 10,000 coins |
| 20,000 | Pile Shrinker | 12,000 coins + permanent +5% coins |
| 25,000 | Heavy Equipment | 15,000 coins; Extractor unlocks |
| 35,000 | Good Behaviour | 25,000 coins + "Good Behaviour" title |

| Items sorted | Milestone | Reward |
|---|---|---|
| 50,000 | Halfway Out | 50,000 coins + permanent +10% coins |
| 60,000 | Parole Hearing | 60,000 coins |
| 75,000 | Trash Legend | "Trash Legend" title + Rainbow trail |
| 90,000 | Light at the Gate | 100,000 coins + permanent +10% coins |
| 100,000 | FREEDOM | "Escaped" title, gate opens |

## Challenges (GOALS panel)

20 challenges in 7 tracks. Each tier shows a banner and pays out when reached; the third tier of most tracks also gives a title.

| Track | Counts | Tier I | Tier II | Tier III |
|---|---|---|---|---|
| Recycler | Items in the Recyclable bin | 100 → 300 coins | 1,000 → 3,000 | 10,000 → 20,000 + title |
| Composter | Items in the Biodegradable bin | 100 → 300 | 1,000 → 3,000 | 10,000 → 20,000 + title |
| Waste Manager | Items in the Non-Biodegradable bin | 100 → 300 | 1,000 → 3,000 | 10,000 → 20,000 + title |
| Gold | Golden Cans binned | 1 → 500 | 10 → 5,000 + Gold trail | 50 → 25,000 + title |
| Throwing | Items thrown into the right bin | 10 → 200 | 100 → 2,000 | 1,000 → 15,000 + title |
| Coins | Coins earned from sorting | 10,000 → 1,000 | 250,000 → 20,000 | 2,000,000 → +10% coins + title |
| Tools | Tools owned | 3 → 1,500 | 7 → 50,000 + Electric trail | — |

## Daily reward

- One claim every 20 hours from the DAILY button: 200 coins + 250 per milestone reached.

- A streak adds +10% per consecutive day, up to +70%; missing more than 48 hours resets it.

### Save slots

- 3 save slots per account. Each slot has its own pile, coins, upgrades and progress.
- A new game always starts at 9 AM in-game time with a fresh 100,000-item pile.

## 6. Economy and upgrades

Coins are the only currency. They come from correct deposits, golden cans, milestones, challenges and the daily reward, and they are spent in the SHOP on carrying capacity, tools, upgrades and perks.

### Trash Sack (gate to everything else)

- Price 150 coins, unlocks at 50 items sorted.
- Raises carrying from 1 item to 3.
- Required before any tool beyond Bare Hands can be bought or equipped, and before Bigger Bag can be levelled.

### Tools

Each tool grabs a cluster of nearby items in one pickup (within a 7-stud cluster radius), capped by free sack space.

| Tool | Price (coins) | Items per grab | Reach (studs) | Pickup speed | Special ability |
|---|---|---|---|---|---|
| Bare Hands | Free | 1 | 9 | 0.8x | — |
| Basic Grabber | 150 (or the 100-item milestone) | 3 | 11 | 1.1x | — |
| Heavy-Duty Grabber | 1,200 | 4 | 12 | 1.5x | — |
| Industrial Trash Claw | 9,000 | 6 | 14 | 2.0x | — |
| Electric Collection Tool | 20,000 | 8 | 16 | 2.7x | Magnet: hold to pull cans from a 40° cone |
| High-Tech Vacuum | 60,000 | 10 | 18 | 3.6x | Vacuum: hold to suck up any trash in front |

| Tool | Price (coins) | Items per grab | Reach (studs) | Pickup speed | Special ability |
|---|---|---|---|---|---|
| Advanced Waste Extractor | 150,000, after 25,000 items | 12 | 22 | 5.0x | Stronger vacuum, tractor beam (8 items, 3 s cooldown) and gold radar |

Tools are bought in order, and every tool after Bare Hands needs the Trash Sack first. Abilities are used by holding left mouse or F (or the on-screen button on phones).

## Levelled upgrades

Each level costs more than the last: price = base × growth ^ level.

| Upgrade | Effect, first to max level | Levels | First price | Max-level price | Total to max |
|---|---|---|---|---|---|
| Bigger Bag | Carry 3 → 400 items (Plastic bag to Mini dumpster) | 13 after the sack | 43 | 24,761 | 60,075 |
| Movement Speed | Walk speed 16 → 30 | 9 | 80 | 13,587 | 28,596 |
| Collection Radius | +0 → +15 studs reach (also boosts abilities) | 8 | 450 | 68,468 | 133,246 |
| Coin Multiplier | 1.0x → 2.0x coins | 10 | 300 | 294,488 | 550,303 |
| Sorting Efficiency | +0% → +40% coins on deposits, shorter deposit cooldown | 6 | 600 | 43,002 | 74,412 |

## Late-game perks

| Perk | Effect | Price | Unlocks at |
|---|---|---|---|
| Auto-Deposit | Deposits for you without walking to the bins | 175,000 coins | 35,000 items |
| Auto-Sort Sack | The sack sorts itself, no trip to the bins needed | 450,000 coins | 50,000 items |

## Pickup speed

Every item takes the same hold time for a given tool: 0.6 s at base, reduced by tools and boosts to a 0.12 s floor. This keeps pickups feeling consistent across item types.

# 7. Monetization

The game sells 8 permanent passes and 9 repeatable products, all linked to live Roblox IDs. Every purchase saves time or adds comfort; none is required to finish, and there are no loot boxes, timers that punish or pay-to-win walls.

## Game passes (one-time)

| Pass | Price (R$) | Effect | Pass ID |
|---|---|---|---|
| All Tools | 499 | Unlocks every tool, including the Extractor | 2055466853 |
| 2x Coins | 299 | Double coins from every correct deposit | 2052916991 |
| Auto-Sorter | 249 | Any bin accepts all trash | 2055502861 |
| VIP | 199 | VIP name tag + 10% more coins | 2054224944 |
| Fast Hands | 199 | Pick up 1.5x faster | 2052382946 |
| +50% Capacity | 149 | Carry 50% more | 2053726949 |
| Gold Radar | 149 | Golden cans glow through the pile | 2053096938 |
| Bin Teleport | 99 | Button to jump between pile and bins | 2052448970 |

## Developer products (repeatable)

| Product | Price (R$) | Effect | Product ID |
|---|---|---|---|
| Crate of Coins | 399 | +100,000 coins | 3721601231 |
| Bag of Coins | 99 | +15,000 coins | 3721601146 |
| Next Tool | 99 | Unlock and equip the next tool (20,000 coins if all owned) | 3721601683 |
| Cleanup Crew | 99 | Clears 500 items, counts toward escape | 3721601873 |
| 2x Coins (15 min) | 49 | Temporary coin boost | 3721601394 |
| Sack Upgrade | 49 | +1 sack level (5,000 coins at max) | 3721601762 |

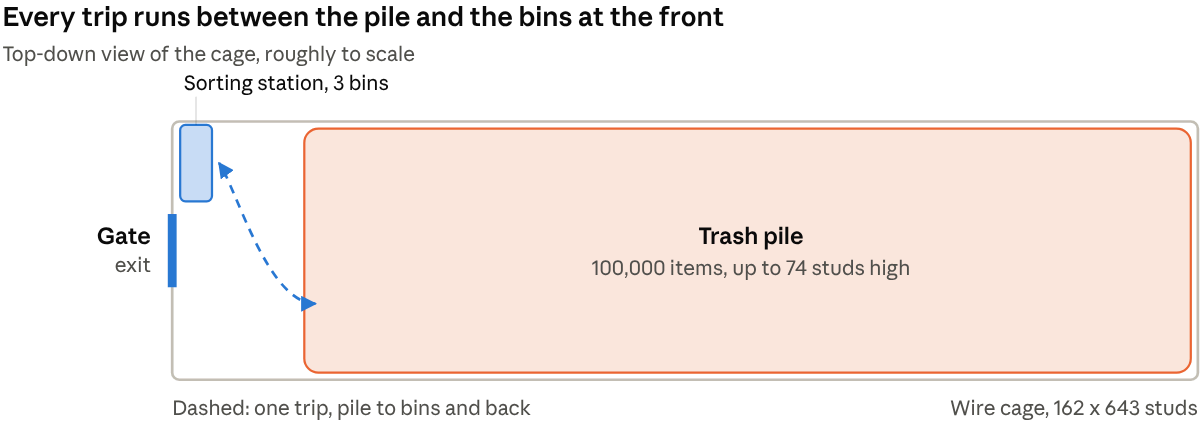
| Product | Price (R$) | Effect | Product ID |
|---|---|---|---|
| Pick-up Boost (15 min) | 39 | 1.5x pickup speed | 3721601483 |
| Speed Boost (15 min) | 29 | +6 walk speed | 3721601606 |
| Pile of Coins | 25 | +2,500 coins | 3721600994 |

### Rules

- Purchases are processed on the server, and each receipt is granted exactly once even if Roblox retries it.
- Managed (regional) pricing is off, so the price shown in the shop always matches the price charged.
- Shop items with no ID show a "not linked yet" message instead of a purchase prompt.
- Open decision: whether Sack Upgrade and All Tools should still respect the 50-item sack unlock.

## 8. World and level design

The whole game happens in one long cage: the gate and sorting station at the front, the pile filling the rest. Every trip is a walk from the pile back to the bins; the sack, tools, Bin Teleport and the late-game perks exist to cut that walk down.




cage layout · top-down

The gate and the bins share the front wall, so every sorted item ends at the same spot the player will finally walk out of.

## Areas

| Area | Size / position | Purpose |
|---|---|---|
| Wire cage | 162 × 643 studs, 82 studs high | Play space; mesh walls and roof keep the player in but show the outside |
| Trash pile | Fills the cage from about 83 to 661 studs back, up to 74 studs high | The 100,000 items |
| Sorting station | Beside the gate, right side | Blue, green and red bins under a "Which bin?" board |
| Escape gate | Front wall, 46 studs wide | Opens when the sentence is served; progress board above it |
| Rules board | Left of the gate | Six house rules for new inmates |
| Outside | Desert, road, power lines, water tower | Visual only; seen in the intro and through the mesh |



## Art direction

- Bright, stylized, low-poly. Flat-color SmoothPlastic surfaces; no realistic textures, so the ground matches the trash and props.
- Trash colors are bold and readable at a distance so categories can be guessed by shape and color.



## Day and night

- A continuous day/night cycle. Nights are kept bright enough to keep playing.
- Every new game starts at 9 AM.

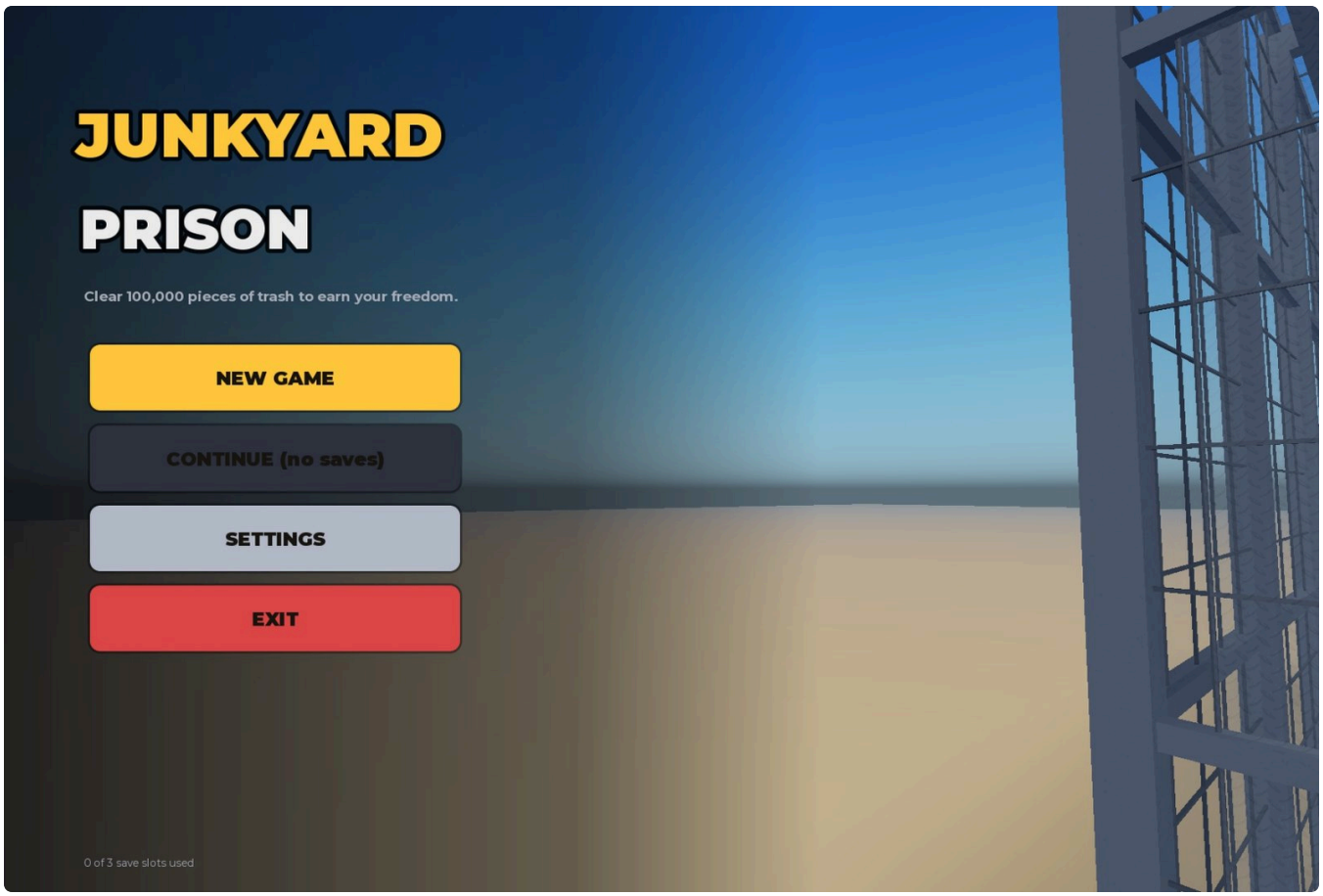


## 9. Onboarding, UI and audio

A new player goes from the main menu to their first deposit in a few minutes: a short intro cutscene sets up the story, then the Warden walks them through the first trip.

### First-time flow

1. **Main menu:** New Game, Continue, Settings, Exit over a slow orbit of the prison; shows how many of the 3 save slots are used.
2. **Loading screen:** tips, including "clear 50 trash to unlock the Trash Sack".
3. **Intro cutscene (new game only):** a prisoner bus drives through the desert, enters the gate, drops the player off, drives out, and the gate slams shut.
4. **Warden tutorial:** PA-style dialog box at the bottom (LIVE • WARDEN • PRISON PA) with Next and Skip. Explains the sentence, picking up, the three bins, coins and the sack.
5. **Free play:** HUD appears; first milestone lands at 25 items.

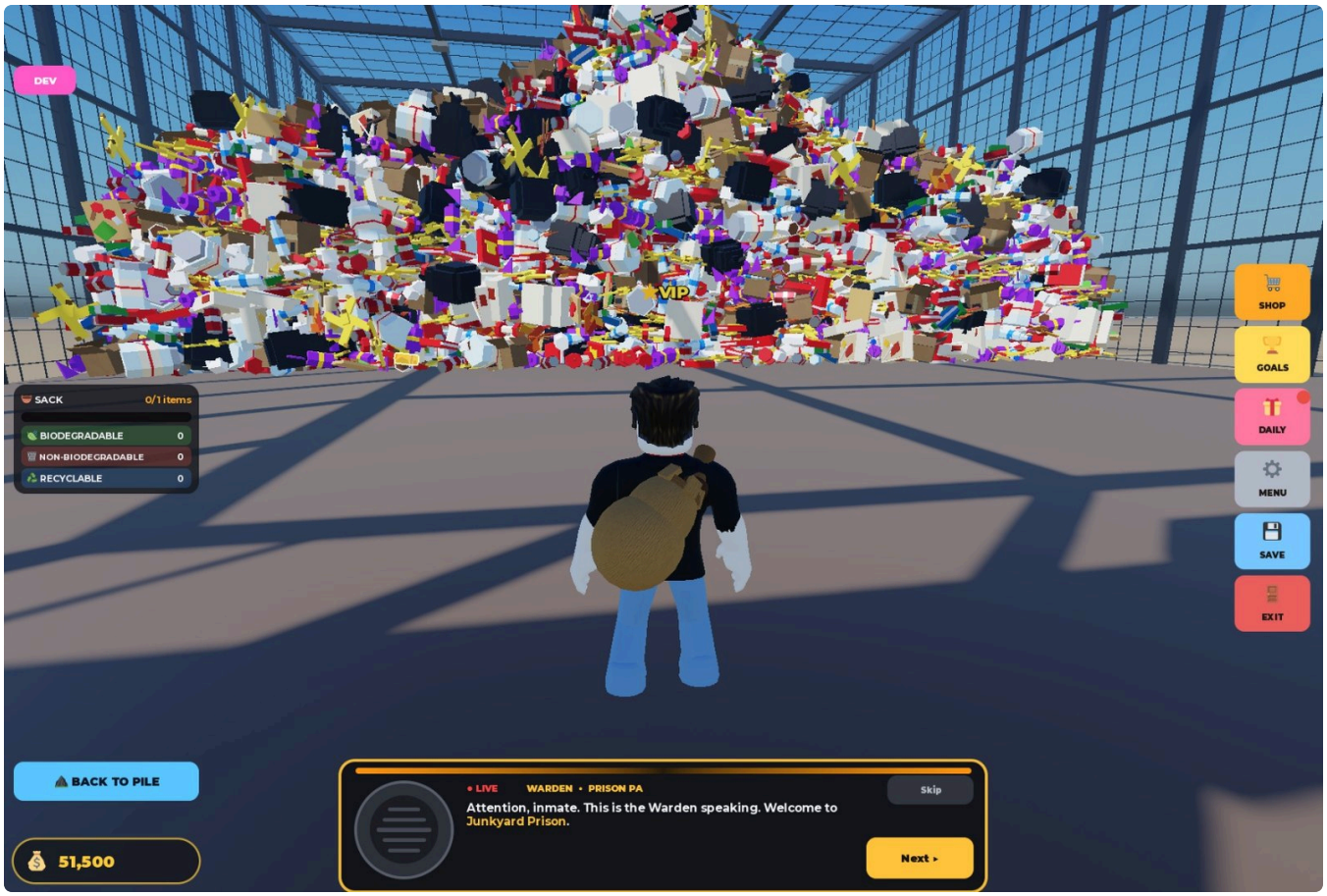

JUNKYARD
PRISON
Clear 100,000 pieces of trash to earn your freedom.
NEW GAME
CONTINUE (no saves)
SETTINGS
EXIT
0 of 3 save slots used


SORTING STATION
Plastic bottles
Cans
Glass jars
Cardboard boxes
Banana peels
Apple cores
Leaves
Food scraps
Paper towels
Wrappers
Chip bags
Foam cups
Garbage bags
Dirty packaging
VIP
RECYCLABLE

# HUD

| Element | Position | Shows |
|---|---|---|
| Sack panel | Left | Items carried / capacity, split by the three categories |
| Coin counter | Bottom left | Current coins |
| Bin Teleport | Bottom left, above coins | Pass owners only: jumps between pile and bins |
| Menu stack | Right | SHOP, GOALS, DAILY (with notification dot), MENU, SAVE, EXIT |
| Pickup bar | Over the target item | Hold progress for the current pickup |
| Toasts and banners | Top / center | Errors, milestones, challenge completions |

The HUD hides during the intro. EXIT asks for confirmation, saves and returns to the main menu.



## Phone and tablet layout

Touch-only devices get their own layout so nothing sits on the thumbstick or jump button; PC is unchanged.

- Menu buttons move into one row along the bottom centre, between the thumbstick and jump button.
- Coins move to the top right, with boosts and the Bin Teleport button under them.
- The sack panel moves to the top left, and the Warden box sits above the button row.
- The game is locked to landscape (either way round); it never rotates to portrait.



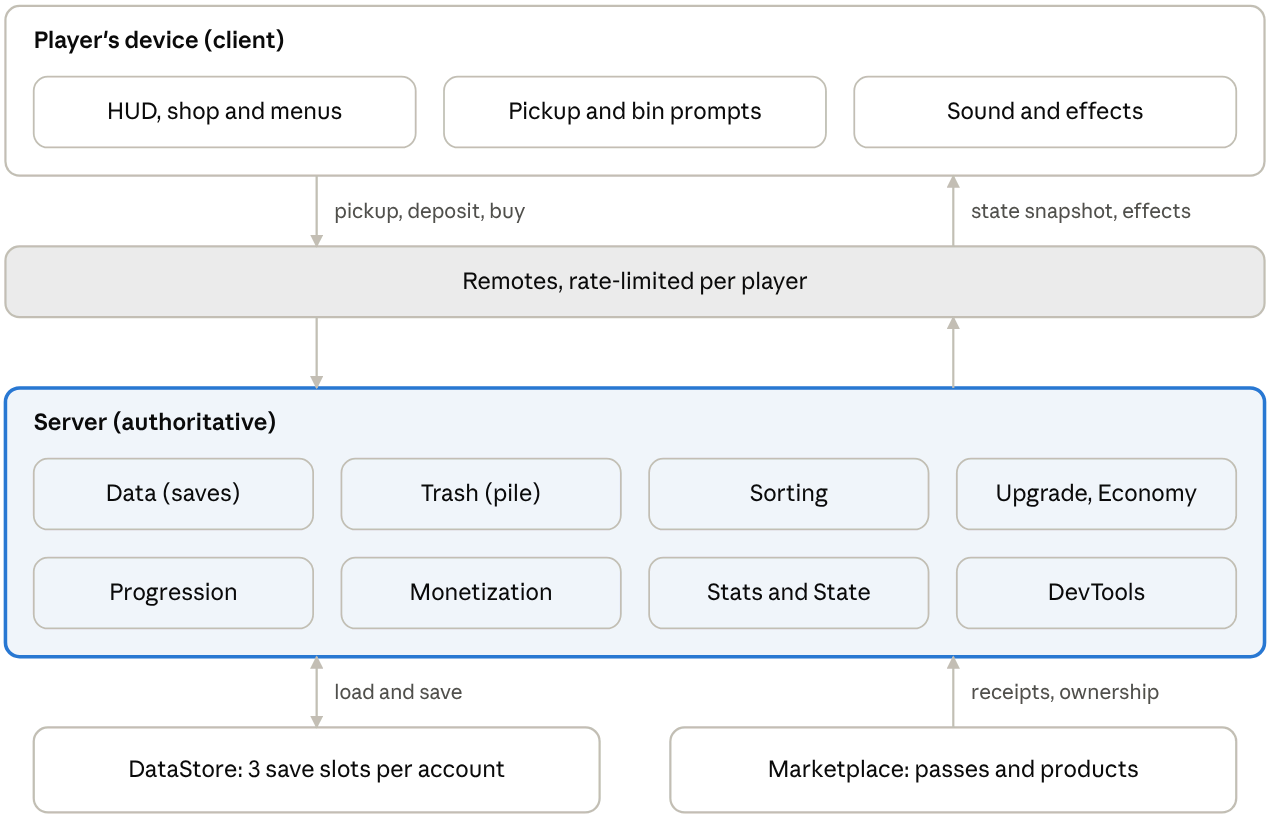

## Audio

- **Music:** one menu track and two in-game tracks, with a Music toggle in settings.
- **Sound effects:** pickup, deposit (plays instantly on press), coins, error, cash-register purchase, milestone, gate, button click, Warden PA, and intro sounds (bus engine, brake, door, gate open and clang, thud).
- All audio uses verified Roblox library assets.

# 10. Technical design

The server owns every rule: what was picked up, what was sorted, coins, purchases and saves. The client only shows state and asks for actions, so a modified client cannot add coins or items.

**The client only asks; the server decides and saves**




architecture · client, remotes, server, Roblox services

Every request crosses the rate-limited remotes; only the server writes saves and grants purchases.

## Server services

| Service | Responsibility |
|---|---|
| Data | Account and slot saves, 3 slots, autosave, slot start/end |
| Trash | The 100,000-item pile, pickups, loose items, exact count |

| Service | Responsibility |
| --- | --- |
| Sorting | Deposits, right/wrong bin, Auto-Sort Sack loop |
| Economy / Upgrade | Coins, Trash Sack, tools, levelled upgrades, perks |
| Progression | Milestones, challenges, daily reward, escape |
| Monetization | Pass ownership, product receipts |
| Stats | Final carry, reach, grab count, pickup time per player |
| State | Sends one snapshot of the player's state to the client |
| DayNight, Cosmetics, ToolAbility | Clock, trails and sack visuals, multi-grab |
| DevTools | Test commands for Studio, the creator and listed dev IDs only |

## Rules that keep it safe

- Every remote is rate-limited per player.
- A pickup is accepted only after the full hold time has passed on the server (with a small tolerance) and the item is within reach.
- Developer-product receipts are recorded on the account, so a retried receipt is never granted twice.
- A save is never overwritten with default data if loading failed.

## Saves

- The pile is saved as a bitset of which of the 100,000 items are gone, plus a list of loose items with positions.
- On load, if the pile does not add up to exactly 100,000 it is rebuilt before the save is applied.

## Performance

- 100,000 single-part, low-poly items: about 10 million triangles, down from 56 million with the original models. All items stay drawn, with no culling.
- Items are anchored until disturbed, so physics only runs on the few that are moving.

- Pickup lookups use a cell-based height map that updates as items are removed, not a full scan.
- Instance streaming is on. Max players per server is 1.

### Mobile

- All actions work with on-screen hold prompts and HUD buttons.
- Touch-only devices get their own HUD layout (section 9); PC keeps the original layout.
- Screen orientation is locked to landscape.
- Tested on a real phone with all 100,000 items drawn: no lag.

## 11. Launch checklist and risks

The game is feature-complete for launch; what remains is the escape badge, two design decisions and publishing.

### Launch checklist

- [x] ~~All 8 passes and 9 developer products created and linked by ID~~
- [x] ~~Managed pricing disabled on every item~~
- [x] ~~Maximum Visitor Count set to 1, social slots disabled~~
- [x] ~~Game icon (512 × 512) and 5 thumbnails (1920 × 1080) made~~
- [x] ~~Tested on a real phone; touch layout and landscape lock added~~
- [x] ~~Tool prices, upgrade table and milestone list filled in from the game config~~
- [ ] Create the escape badge and add its ID to the config
- [ ] Decide whether dev tools stay on for the creator in the live game
- [ ] Decide whether Sack Upgrade and All Tools respect the 50-item sack unlock
- [ ] Publish the latest version, answer the maturity question, launch to 16+
- [ ] Later: full maturity questionnaire, 2-Step Verification and review for the 9+ audience

### Risks

| Risk | Impact | Mitigation |
|---|---|---|
| Low-end phones struggle with 100,000 drawn items | Low frame rate, early quits | Tested on a real phone with no lag; low-poly models cut triangles about 80% |
| 100,000 items feels too long | Players drop off mid-game | Tool and perk curve, 18 milestones, 20 challenges, daily reward |
| Players expect multiplayer | Confusion with 1-player servers | Say "single-player" in the description and thumbnails |
| Data loss on a failed load | Lost progress, bad reviews | Saves are never overwritten by defaults; 3 slots per account |

---

# Part II — Detailed design

Part I above is the summary. Part II specifies every system in full, for implementation, review and QA.

## 12. Feature specifications

Every rule below is enforced on the server; the client only shows prompts and asks. Numbers come from the game's config at the time of writing.

### 12.1 Pick up

- **One prompt, moved to the best target.** A single hold prompt is re-placed every 0.1 s. Target = the item under the mouse (PC) or screen centre (touch, gamepad) if in reach; otherwise the nearest item by surface distance, favouring items in front of the camera and on screen.
- **Hold time** = 0.6 s ÷ pickup speed (tool × Fast Hands × Pick-up Boost), never below 0.12 s. Every item type takes the same time.
- **Server checks on finish:** held for at least 75% of the hold time (latency allowance), item still in the world and unclaimed, within reach (tool reach + Collection Radius + 2 studs, plus 8 studs of slack for movement), and free space in the sack.

- **Claim is atomic.** Nothing yields between the checks and the removal, so an item can never be taken twice.
- **Refusal** resets the hold bar on the client at once and shows why ("Your hands are full!", "Your bag is full!").
- **Physics:** up to 30 anchored items resting within 7 studs above the taken one wake and slide, then re-freeze once they settle.

## 12.2 Multi-grab

- A tool with Grab N also takes the N−1 nearest live items within 7 studs of the picked one, capped by free sack space.
- The prompt shows the real count: "Pick up (x3)". Collection Radius adds reach only; it never adds items per grab.

## 12.3 Carrying

- Capacity: hands 1 → Trash Sack 3 → Bigger Bag levels up to 400. The +50% Capacity pass multiplies the result (rounded down).
- The inventory is stored per item type; the HUD sack panel shows totals per bin.
- The sack on the player's back grows with capacity in 12 visual size steps.

## 12.4 Deposit

- Each bin has an instant prompt (no hold) usable within 12 studs; the server accepts within 16.
- One deposit moves **every carried item of that bin's category**. Wrong items stay in the sack and a toast lists what is left.
- Wrong-only deposit: nothing moves, and the message names the right bin and its colour.
- Cooldown between deposits: 1.0 s, falling to 0.25 s with Sorting Efficiency.
- **Payout per item** = base coins × Coin Multiplier level × (1 + permanent bonus % + rebirth % + VIP 10%) × 2x Coins pass × 2x boost × (1 + Sorting %), rounded.
- The deposit sound plays on the client the moment the prompt fires, before the server answers.

## 12.5 Throwing

- G, gamepad B or the on-screen Throw button throws the last item picked up.

- Facing a bin within 30 studs, the arc is solved to land inside it (11-stud arc). Otherwise it is a forward lob.
- Right bin: counts as sorted, pays normally and adds to the Throwing challenge. Wrong bin: it pops back out and the message names the right bin.
- Thrown items that miss stay in the world as loose items and are saved. Beyond 250 loose items the oldest stop simulating; none are ever deleted.

## 12.6 Tool abilities

Hold left mouse, F or the on-screen button. The server picks the real items; sack space still limits.

| Tool | Ability | Cone | Length | Rate |
|---|---|---|---|---|
| Electric Collection Tool | Magnet: cans only (aluminium, crushed, golden) | 40° | 16 studs | 5 items/s |
| High-Tech Vacuum | Vacuum: any trash | 35° | 16 studs | 4 items/s |
| Advanced Waste Extractor | Stronger vacuum + tractor beam + gold radar | 45° | 20 studs | 7 items/s |

- **Tractor beam** (Q, gamepad L2, BEAM button): lifts up to 8 items clumped within 6 studs of the aim point, up to 40 studs away, every 3 s.
- **Collection Radius** also boosts abilities: +1 stud cone length per stud, beam range +70% and clump +40% of the bonus, beam items + half the tool's extra grab.

## 12.7 Perks

- **Auto-Deposit:** within 10 studs of a bin, matching items drop in by themselves (checked every 0.25 s). A glowing ring on the ground marks each bin's zone.
- **Auto-Sort Sack:** every 0.25 s the whole sack is sorted wherever the player stands, paid like a normal deposit; no range or cooldown.

## 12.8 Passes and boosts

| Item | Exact effect |
|---|---|
| VIP | +10% coin bonus; VIP name tag over the head |

| Item | Exact effect |
|---|---|
| 2x Coins | ×2 on every sorting payout |
| +50% Capacity | Capacity × 1.5; also shows the sack panel before the sack is bought |
| Auto-Sorter | Any bin accepts all carried trash (manual and Auto-Deposit; not Auto-Sort Sack) |
| All Tools | Marks every tool owned; equipping past Bare Hands still needs the sack |
| Fast Hands | Pickup speed ×1.5 |
| Gold Radar | Nearest golden can glows through the pile with a distance label |
| Bin Teleport | Button: to the bins, or back to the last spot on the pile; 2 s cooldown |
| Boosts (15 min) | Coins ×2, walk speed +6, or pickup ×1.5; buying again extends from the current expiry |

## 12.9 Escape and rebirth

- At 100,000: Escaped flag set, fastest-escape time recorded, badge awarded (once its ID is set), save forced, YOU ESCAPED! screen with WALK OUT and KEEP PLAYING.
- **Rebirth** (MENU, after escaping, tap twice to confirm): resets coins, items, upgrades, tools, milestones, challenges, counters and the pile. Keeps titles, trails and Robux passes. Each rebirth adds +25% coins permanently.

## 12.10 Daily reward

- Claimable every 20 hours. Coins = (200 + 250 × milestones reached) × (1 + streak bonus).
- Streak bonus = +10% per consecutive claim after the first, capped at +70%; more than 48 hours between claims resets it.

## 12.11 Saves, slots and session

- 3 save slots per account. New Game needs an empty slot; Continue needs a used one; a slot can be emptied unless it is the one being played.

- Auto-save every 1, 2 (default), 5 or 10 minutes, or off. Manual SAVE at most once per 6 s. Always saves on leave, on EXIT and on server shutdown.
- A run timer counts play time until escape.

### 12.12 Day and night

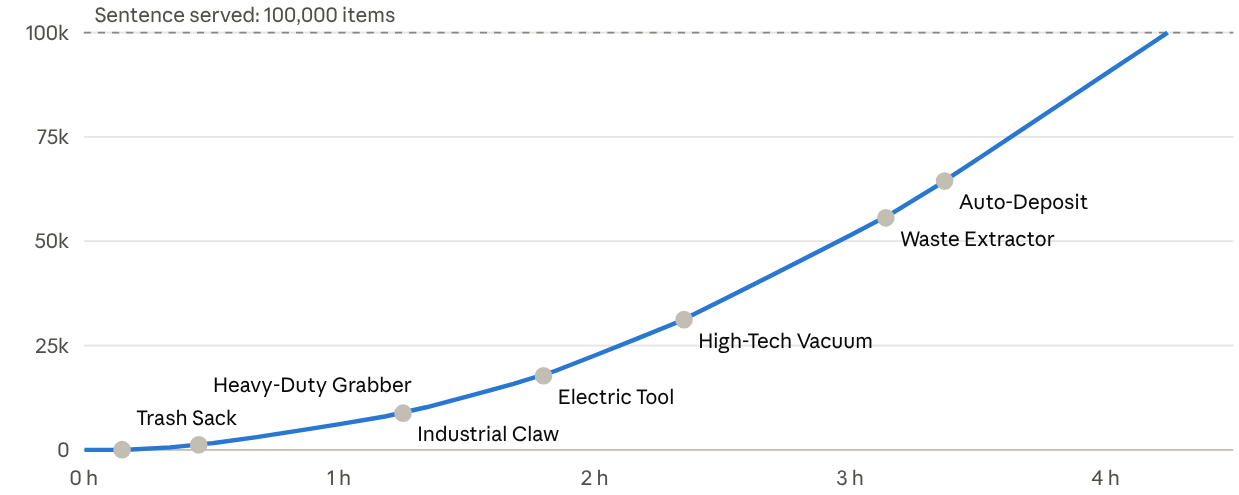
- One in-game day lasts 12 real minutes. Floodlights switch on at 18:00 and off at 06:00.
- Every new game starts at 09:00.

## 13. Economy and balancing

A simple model of the current numbers says a focused player reaches 100,000 in about 4.2 hours of non-stop play, earning about 1.1 million coins from sorting. Real sessions with breaks and wasted trips will run longer; the model is for comparing changes, not a promise.

**A focused player escapes in about 4 hours of play**

Modelled items sorted vs hours of non-stop play; 100,000 reached at 4.2 h. Dots = when each unlock becomes affordable.




Economy model run on the game config, 23 Oct 2026 · idealised play, no breaks

### Model assumptions

- Average one-way walk between the pile and the bins: 70 studs, at the current walk speed.
- 0.8 s to line up each pickup, plus the hold time; a held vacuum is used when it is faster than prompt pickups.
- Up to 3 deposit prompts per trip, at the deposit cooldown or 1 s, whichever is longer.

- Average base value of an item: 4.17 coins (weighted by how often each type appears, golden cans included).
- The player always buys the cheapest available upgrade, tool or perk as soon as it is affordable. Milestone coins and permanent bonuses are counted; challenge and daily rewards are not.

### Coin sinks

| Sink | Total coins to own all |
|---|---|
| Trash Sack + 6 tools | 240,500 |
| All levelled upgrades to max | 846,632 |
| Both perks | 625,000 |
| **Everything** | **1,712,132** |

### Findings and tuning levers

- **Pacing is fast for a 100,000 goal.** About 4 hours of perfect play. To stretch it, lower item coin values or the Coin Multiplier curve before touching item counts.
- **The Waste Extractor arrives late.** It unlocks at 25,000 items but is affordable at about 55,600. Fine as a late treat; lower to ~100,000 coins to make it a midgame goal.
- **Auto-Sort Sack is never bought before escape.** At 450,000 coins it competes with maxing the Coin Multiplier (550,303 total). Either drop it to about 250,000, or keep it as a post-escape and rebirth goal on purpose.
- **The first 50 items take about 9 minutes by hand.** That is the intended "grind before the sack". Watch first-session drop-off here closely.
- **Robux value check:** the Crate of Coins (100,000 coins for R$399) equals about 6% of everything purchasable, which keeps it a convenience, not a skip.

Validate these numbers in playtests: log time to 50, 1,000 and 10,000 items (section 18).

## 14. UI screen specifications

All UI is built in code and scaled with one UIScale from a 1100 × 640 reference, so it fits phones through 4K. Only one panel is open at a time; Esc or the red X closes it, every panel opens with a short pop, and every button plays the click sound.

## 14.1 Main menu and loading

| Element | Behaviour |
|---|---|
| Title + tagline | JUNKYARD / PRISON, "Clear 100,000 pieces of trash to earn your freedom." over a slow camera orbit of the prison |
| NEW GAME | Starts a fresh sentence in an empty slot; plays the intro cutscene |
| CONTINUE | Resumes a used slot; greyed with "(no saves)" when all slots are empty |
| SETTINGS | Sound, music and auto-save options (shared by all slots) |
| EXIT | Leaves the game |
| Footer | "N of 3 save slots used" |
| Loading screen | Game title, status line ("Connecting…"), rotating tips; the character spawns only after the world has loaded |

## 14.2 Shop (SHOP button)

A 640 × 440 panel with two tabs.

| Tab / section | Rows | Button states |
|---|---|---|
| COINS › UPGRADES | 5 levelled upgrades: name, "Level x / max", current → next value | Green price if affordable, grey if not, MAXED, 🔒 Need sack (Bigger Bag before the sack) |
| COINS › TOOLS | Trash Sack row, then 7 tools with speed, reach and grab line; ability line for the top 3 | Sack: 🔒 n/50 trash → price → ✅ OWNED. Tools: EQUIPPED, EQUIP, 🔒 Need sack, 🔒 Buy previous, 🔒 25K trash, price |
| COINS › HIGH-TIER PERKS | Auto-Deposit, Auto-Sort Sack with description | 🔒 35K / 50K trash, price, ✅ OWNED |

| Tab / section | Rows | Button states |
|---|---|---|
| ROBUX › PERMANENT PASSES | 8 passes: name, description | R$ price (opens Roblox purchase prompt) or ✅ OWNED |
| ROBUX › COINS & BOOSTS | 9 products; Next Tool names the tool it gives | R$ price; disabled when nothing is left to give (all tools owned, sack maxed) |
| Footer | "Optional extras - everything can also be earned by playing." | — |

### 14.3 Goals (GOALS button)

- Subtitle: "Challenges + escape milestones. Rewards unlock automatically."
- ⭐ CHALLENGES (done / 20): unfinished first; each row shows name, target, reward and a progress bar with "have / target".
- 🏁 ESCAPE MILESTONES: all 18 with reward text and a progress bar; finished rows turn green with a tick.

### 14.4 Daily (DAILY button)

- 440 × 260 panel. Ready: "Your daily coins are ready!" and a green CLAIM button. Not ready: "Next reward in Xh Ym" and a grey COME BACK LATER.
- A red dot on the DAILY button shows when a claim is ready.

### 14.5 Menu (MENU button)

| Row | Control |
|---|---|
| Sound effects | ON / OFF |
| Music | ON / OFF |
| Auto-save | ON / OFF |
| Auto-save every | 1 / 2 / 5 / 10 min, plus "Slot N • last saved …" |
| How to play | PLAY replays the Warden tutorial |

| Row | Control |
|---|---|
| Rebirth (count) | Locked until escape; REBIRTH needs a second tap within 3 s |
| Titles and trails | One row per earned item with EQUIP / UNEQUIP |

### 14.6 Pop-ups, banners and toasts

- **Confirm pop-up** (520 × 250): title, body, CANCEL and a red YES. Used by EXIT ("save and return to menu").
- **Banner** (centre): milestone (🏆 name, "N trash cleared - reward unlocked", 5 s) and challenge (⭐ name, 4 s).
- **Toasts** (top centre, max 3, 3.5 s): errors red, success green, info blue. Examples: wrong bin, bag full, auto-saved.
- **Escape screen**: full-screen dim, "🎉 YOU ESCAPED! 🎉", explanation, WALK OUT (guides the player to the open gate) and KEEP PLAYING.

### 14.7 In-world UI

- **Pickup prompt:** "Pick up" / "Pick up (xN)", object text "Item name • category".
- **Bin prompts:** "Put in bin", object text "♻️ Recyclable bin" etc.
- **Progress board** above the gate: sentence counter and run timer.
- **Floating text** over bins and pickups for coins earned; a bouncing arrow marks the tutorial target.
- **Gold radar label:** "GOLD 23m" over the nearest golden can.
- **DEV button and panel** (F2): developers only (Studio, the creator, listed IDs); every command is re-checked on the server.

## 15. Narrative content

All player-facing story text in the game, word for word.

## 15.1 Intro cutscene (new game only, 19.4 s, skippable)

| Time (s) | Shot and action | Caption |
|---|---|---|
| 0.0 | Fade in from black; bus drives up the desert road, engine loop | — |
| 0.6 | Bus approaching | "Inmate transport arriving at Junkyard Prison..." |
| 4.4 | Air brakes as the bus stops outside | — |
| 5.0 | Gate swings open (creak) | "The gate opens." |
| 7.4–10.4 | Bus pulls into the yard and parks; door hisses open | — |
| 11.0 | The inmate is thrown out of the door | **"OUT YOU GO!"** |
| 12.1 | Lands in the dirt (thud, camera shake), stands up | — |
| 12.8–13.3 | Door closes; bus reverses out | — |
| 16.0 | Gate starts to close | **"SENTENCE: 100,000 PIECES OF TRASH"** / "The gate won't open again until you've cleared them all." |
| 18.2 | Gate slams (clang, camera shake) | — |
| 19.4 | Fade out; HUD and controls return; the Warden begins | — |

## 15.2 Warden tutorial script

Each line types out over the PA with a radio squelch. A bouncing arrow marks the target of each step; task steps advance by themselves when done.

| Step | Warden line | Advances when |
|---|---|---|
| 1a | "**Attention, inmate.** This is the Warden speaking. Welcome to **Junkyard Prison**." | Next |

| Step | Warden line | Advances when |
|---|---|---|
| 1b | "You were caught **littering**. Again. And again. You ignored every **warning** and dodged every **penalty** we sent you." | Next |
| 1c | "So here's your sentence: clear **100,000** pieces of trash from this landfill. Every. Single. Piece." | Next |
| 1d | "Clear them all and that gate opens - you walk out free. Until then, you sort. **Get to work.**" | Let's go |
| 2 | "First job: walk up to the trash mountain and **hold E** (or hold the button) on any piece to pick it up." | First pickup |
| 3 | "You've got: **[item]**. That goes in **[bin]**. Carry it to that bin and **hold E**. Wrong bin and it comes right back to you." | First correct deposit |
| 4 | "Every correct bin pays coins - spend them in the 🛒 **SHOP**. For now it's one piece at a time, by hand. Clear **50 trash** and a **Trash Sack** unlocks in SHOP > TOOLS." (SHOP button pulses) | Got it |
| 5 | "Your sentence counter is on the big screen above the gate. Spend your coins in the SHOP to clean faster. **The Warden is watching.**" | Start |

Skip ends the tutorial at any step. Progress is saved, so returning players never see it again; MENU › How to play replays it.

## 15.3 System messages

| Situation | Text |
|---|---|
| Wrong bin (deposit) | "Wrong bin! This is the [bin] hole. Your [item] goes in the [bin] ([COLOUR]) hole." |
| Wrong bin (throw) | "Wrong bin! [item] goes in the [bin] ([COLOUR]) bin." |
| Mixed sack | "Still carrying: N [bin] ([COLOUR]), …" |

| Situation | Text |
|---|---|
| Nothing carried | "You're not holding any trash - pick some up from the mountain first!" |
| Hands full | "Your hands are full! Put it in the right bin first. (A Trash Sack unlocks in SHOP > TOOLS after 50 trash.)" |
| Bag full | "Your bag is full! Put your trash in the right bins." |
| Tool before sack | "Get the Trash Sack first (SHOP > TOOLS)" |
| Daily claimed | "Daily reward: +N coins (streak S)" |
| Rebirth | "Rebirth N! +X% coins forever" |
| Auto-save | "💾 Auto-saved" |

## 15.4 Titles and trails

| Reward | Kind | Earned from |
|---|---|---|
| Clean Freak | Title | 10,000 items |
| Sparkle | Trail | 10,000 items |
| Good Behaviour | Title | 35,000 items |
| Trash Legend | Title | 75,000 items |
| Rainbow | Trail | 75,000 items |
| Escaped | Title | 100,000 items |
| Recycler, Composter, Waste Manager | Titles | Tier III of each bin challenge |
| Treasure Hunter | Title | 50 golden cans |
| Gold | Trail | 10 golden cans |
| Sharp Shooter | Title | 1,000 items thrown into the right bin |
| Tycoon | Title | 2,000,000 coins earned |
| Electric | Trail | Owning all 7 tools |

The first title and trail earned are equipped automatically; others are switched in MENU.

## 16. Audio list

15 sound cues and 3 music tracks, all public Creator Store or built-in Roblox audio. Silent lead-ins are trimmed with a play region so every cue lands on the action; all clips are preloaded when the game starts.

### 16.1 Sound effects

| Cue | Asset | Volume | Region / pitch | Plays when |
|---|---|---|---|---|
| Collect | rbxassetid://9120258581 (Trash Movement 2, PSE) | 0.45 | 0.15–0.6 s | Pickup succeeds; tool ability pulls items |
| Deposit | rbxassetid://9118193716 (Road Cylinder Impact 1, PSE) | 0.5 | 0.4–1.3 s | Bin prompt pressed (predicted on the client) |
| Coins | rbxassetid://1169806635 (Coin Collect 2) | 0.45 | — | Coins credited |
| Error | rbxasset://sounds/electronicpingshort.wav (built-in) | 0.5 | pitch 0.45 | Wrong bin, bag full, refused action |
| Purchase | rbxassetid://120891770644830 (cash register) | 0.6 | 0.26–1.04 s | Any coin or Robux purchase |
| Milestone | rbxassetid://9045298695 (Crack The Safe, APM) | 0.55 | — | Milestone or challenge banner |

| Cue | Asset | Volume | Region / pitch | Plays when |
|---|---|---|---|---|
| Gate | rbxassetid://9113969124 | 0.6 | — | Escape: gate opens |
| Click | rbxasset://sounds/clickfast.wav (built-in) | 0.35 | — | Every UI button |
| Warden | rbxassetid://9125936640 (Shortwave Bleep Radio, PSE) | 0.35 | 0–0.7 s | Each new Warden line |
| BusEngine | rbxassetid://9112768547 (Diesel Train 1, PSE) | 0.45 | loop | Intro, bus driving |
| BusBrake | rbxassetid://7993386297 | 0.55 | — | Intro, bus stops |
| BusDoor | rbxassetid://7993386297 | 0.45 | pitch 1.5 | Intro, door opens and closes |
| GateOpen | rbxassetid://9113969124 (Creaks And Squeaks 6, PSE) | 0.6 | — | Intro, gate swings |
| GateClang | rbxassetid://9116603727 (Metal Door 3, PSE) | 0.7 | — | Intro, gate slams |
| Thud | rbxassetid://9113469796 (Body Fall Dirt 4, PSE) | 0.6 | 0–1.2 s | Intro, inmate lands |

## 16.2 Music

| Playlist | Track | Asset | Volume |
|---|---|---|---|
| Menu | Pretty Elegance (APM) | rbxassetid://1838930204 | 0.3 |
| In game | Showtime! (APM) | rbxassetid://1842172845 | 0.22 |
| In game | Dig It (APM) | rbxassetid://1840081489 | 0.2 |

Menu music plays on the main menu and loading screen; the in-game playlist takes over once the player spawns. Sound effects and music each have an ON/OFF switch in MENU, saved on the account.

### 16.3 Audio rules

- Only verified, public audio IDs; no uploads with unclear rights.
- Feedback sounds must start within 50 ms of the action: trim silence with a play region, never add delay.
- No more than one music track at a time; music sits well under sound effects (0.2–0.3 vs 0.35–0.7).

## 17. Technical appendix

One save record per account, every request rate-limited, and a 100,000-item physical pile kept exact by index. Values are from the game config at the time of writing.

### 17.1 Save data

- **DataStore** `TrashPrison_PlayerData_v1`, key `u_<UserId>`, data version 1 (missing fields are filled from the template on load).
- **Record:** `Account` = {Slots "1"–"3", SlotSaved times, LastSlot, Settings, Receipts, ReceiptOrder} + `Lock` = {JobId, Time}.
- **Session lock:** a server only writes a record it holds; locks older than 600 s count as stale. Loads retry up to 5 times (3 s apart) while another server still holds the lock.
- **Retries:** every DataStore call tries 4 times with 1, 2 and 4 s back-off. A failed load kicks the player with "Your progress is safe - please rejoin" instead of handing out a blank save.
- **Receipts** live on the account (the last 100), so a purchase is never granted twice, even across slots.

| Slot field | Holds |
|---|---|
| Coins, LifetimeCoins, Units | Wallet, lifetime earnings, items sorted |
| Inventory | Carried items per type |
| Upgrades | Level per levelled upgrade (Capacity 1 = Trash Sack) |

| Slot field | Holds |
|---|---|
| Tools, EquippedTool | Owned tools, current tool |
| Milestones, Challenges, Stats | Reached milestones, finished challenges, challenge counters |
| Titles, Trails, EquippedTitle, EquippedTrail | Cosmetics |
| CoinBonus, Rebirths, Escaped, EscapeCount | Permanent bonus %, rebirth count, escape state |
| Boosts | Expiry time per boost |
| Daily | Last claim time, streak |
| Pile, PileVersion | 100,000-bit map of taken pile items, saved as 25,000 hex characters |
| Loose | Types of items thrown or dropped but not sorted |
| Perks, Tutorial, PlayTime, BestTime | Perks owned, tutorial step, run timer, fastest escape |

## 17.2 Remotes

Every client→server call passes a per-player, per-action token bucket: 12 calls per 2 s. Calls that fail it are dropped (a pickup is also reset on the client).

| Remote | Type | Direction | Purpose / extra checks |
|---|---|---|---|
| Account | Function | C→S | list / new / continue / delete slot, settings, save (once per 6 s), exit |
| ClientReady | Event | C→S | World loaded, spawn the character (only with a slot running) |
| PickupBegin, PickupFinish | Event | C→S | Pick up (section 12.1 checks) |
| DropItem | Event | C→S | Throw the last item picked up |
| ToolUse | Event | C→S | Start, aim, stop a tool ability; fire the beam (cooldown on the server) |

| Remote | Type | Direction | Purpose / extra checks |
|---|---|---|---|
| BuyUpgrade, BuyTool, BuySack, BuyPerk | Event | C→S | Purchases with coins (price, unlocks and sack rule checked) |
| EquipTool, EquipCosmetic | Event | C→S | Equip owned items only |
| ClaimDaily, Rebirth, TutorialStep | Event | C→S | Daily (20 h), rebirth (after escape), tutorial step (+1 only, or done) |
| Teleport | Event | C→S | Bin Teleport pass only, 2 s cooldown |
| DevCommand | Function | C→S | Dev tools; refused for non-developers |
| StateSync, Notify, Effect | Event | S→C | State snapshot, toasts, sounds and visual effects |
| GetState | Function | C→S | First state snapshot |

Bin deposits use proximity prompts handled on the server, so there is no deposit remote to spoof.

## 17.3 The pile

- 100,000 numbered items (attribute PileIndex) in 56 chunk models of 40 × 40 studs, each with streaming-mesh level of detail.
- Items are one part each, anchored at rest, with touch events off (thrown items turn them on for the bin sensors).
- **Waking:** taking an item releases up to 30 resting items above it within 7 studs (8 for tool abilities). The server owns their physics.
- **Settling:** an item still for 1 s freezes if buried more than 2.5 studs; on slopes steeper than about 36° it gets up to 3 nudges downhill; any item moving for 25 s is frozen. Over 1,200 moving items, the lowest resting ones freeze first.
- **Recovery:** an item that falls out of the world is put back in front of the pile, never deleted.
- **Integrity:** at every new game the pile is rebuilt if it is not exactly 100,000; on load, the saved bit map hides taken items and loose items are placed back in front of the pile.

### 17.4 Performance budget

| Item | Value |
|---|---|
| Pile triangles | About 10 million (68–212 per item) |
| Moving items | Soft cap 800, hard cap 1,200 |
| Loose items simulated | 250 newest |
| Height-map refresh | Once per second, only cells with moving items |
| Client target search | Every 0.1 s, at most 250 parts in a 15-stud radius |
| Players per server | 1 |

### 17.5 Build notes

- Code lives in ReplicatedStorage.TrashPrison (Config, Util, Remotes), ServerScriptService.TrashPrisonServer (Main + Services) and StarterPlayerScripts (MainMenu, TrashPrisonClientMain + TrashPrisonClient modules).
- All tuning lives in Config modules: Upgrades, Progression, Monetization, GameSettings, TrashTypes, Categories.
- A Studio-only test hook drives the real services for automated checks; it does not exist in live servers.

## 18. Analytics and KPIs

Launch decisions come from three numbers: how many new players reach the Trash Sack, how many come back the next day, and how far the pile pacing in section 13 holds up. The Creator Dashboard covers retention, sessions and revenue; the onboarding funnel and economy need custom events (AnalyticsService), which are not built yet.

### 18.1 Dashboard metrics to watch

| Metric | Where | Starting target (revise after week 1) |
|---|---|---|
| Day 1 retention | Creator Dashboard › Analytics | 20% or more |

| Metric | Where | Starting target (revise after week 1) |
|---|---|---|
| Day 7 retention | Creator Dashboard › Analytics | 5% or more |
| Average session length | Creator Dashboard › Analytics | 12 minutes or more |
| Payer conversion | Creator Dashboard › Monetization | Track only; no target before data |
| Revenue per pass / product | Creator Dashboard › Monetization | Find the 2–3 items that sell; rethink the rest |
| Client crashes and errors | Creator Dashboard › Monitoring | No new error types after an update |

The targets are internal starting points for a first release, not Roblox benchmarks.

## 18.2 Onboarding funnel (custom events to add)

1. Main menu shown
2. New Game pressed
3. Intro finished (or skipped)
4. Tutorial step 2: first pickup
5. Tutorial step 3: first correct deposit
6. Tutorial finished (or skipped)
7. 50 items sorted
8. Trash Sack bought
9. 1,000 items
10. 10,000 items
11. Escaped

The biggest expected drop is between steps 6 and 8, the 50 items carried one at a time; this is the step to watch after launch.

## 18.3 Economy events (custom, to add)

- **Sources:** sorting, golden cans, milestones, challenges, daily, Robux coin packs.

- **Sinks:** each tool, each upgrade level, each perk, the sack.
- Compare real time-to-unlock against the model in section 13.

### 18.4 Technical health

- Save failures and load kicks (warnings from DataService).
- Frame rate on phones near the pile (Developer Console › Stats in live tests).
- Refused pickups per minute: a sudden rise points to lag or an exploit attempt.

## 19. Accessibility, localization and platforms

The game is readable without sound and without telling colours apart; the main gaps are an option to turn off camera shake and checking gamepad navigation in the panels.

## 19.1 Accessibility

| Area | In the game now | Still to do |
|---|---|---|
| Colour | Every bin and category also has an icon and a text label; messages name the colour in words | Check the red and green bins with a colour-blindness simulator |
| Hearing | All story and tutorial text is on screen; the intro has captions; sound and music can be switched off | — |
| Motor | Hold times are short (0.12–0.6 s); throws auto-aim at bins; Auto-Deposit and Auto-Sort remove trips late in the game | Consider a tap-to-pick-up option |
| Reading | Warden lines are short; panels use icons with every label | Keep new text under about 25 words per line |
| Motion | — | Add a setting to turn off intro camera shake |
| Pace | The intro and tutorial can be skipped; the tutorial can be replayed | — |
| Screen size | One UI scale for every screen; separate phone layout | Check text size on the smallest phones |

## 19.2 Localization

- All text is English and written in code, so Roblox's automatic translation can pick it up; turn it on in the Creator Dashboard at launch.
- Words baked into pictures or 3D (the game icon, the bus side, the sorting board) will not translate; keep these few and short.
- Once analytics show the top player countries, review the automatic translation for the top one or two languages by hand.

### 19.3 Platforms and controls

| Action | PC | Phone / tablet | Gamepad |
|---|---|---|---|
| Pick up / deposit | Hold E / press E | Hold the on-screen prompt | Hold X / press X |
| Throw | G | Throw button | B |
| Tool ability | Hold left mouse or F | USE TOOL button | Hold R2 |
| Tractor beam | Q | BEAM button | L2 |
| Close panel | Esc or X | X | — |
| Menus | Mouse | Tap | To be tested |

- **Phone:** landscape only, own HUD layout, tested on a real phone with all items drawn and no lag.
- **Console:** prompts and abilities have gamepad bindings; menu navigation with a gamepad is untested.
- **VR:** not supported.

### 19.4 Content rating

No blood, weapons, gambling, romance or real-world brands. The only rough moment is slapstick: the inmate is tossed out of the bus. Expected maturity result: Minimal or Mild, to be confirmed by the questionnaire.

## 20. QA test plan

27 test cases cover every system; 8 have passed in Studio or on a phone so far, and the rest are to run before launch. Run the full list again after any change to the pile, saves or purchases. Studio tests use the dev tools (DEV button) to jump ahead.

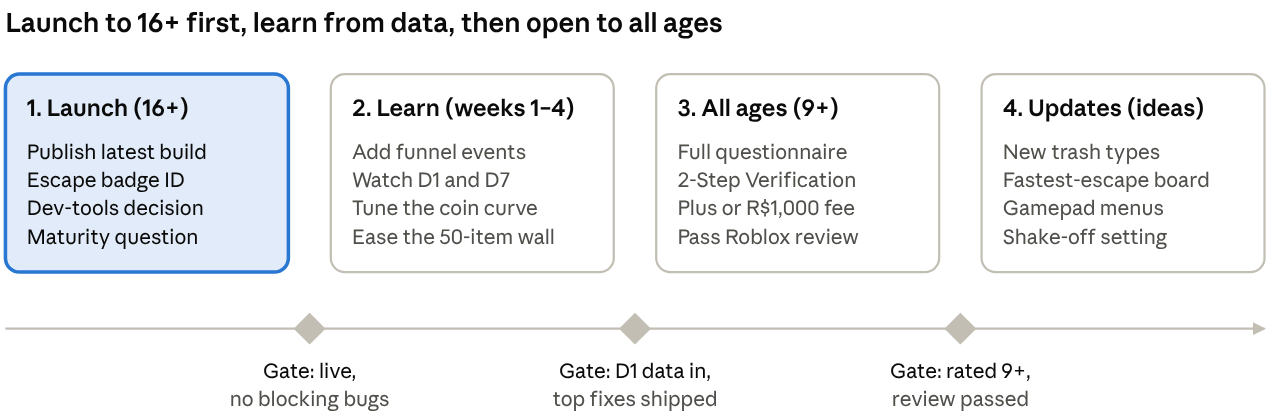
| ID | Area | Test | Expected | Status |
|---|---|---|---|---|
| QA-01 | Pile | Start a new game in any slot | Exactly 100,000 pile items, none duplicated or missing | Passed |
| QA-02 | Saves | Clear some items, drop one, exit, continue | Pile + loose + carried + sorted = 100,000 | Passed |

| ID | Area | Test | Expected | Status |
|---|---|---|---|---|
| QA-03 | World | Start a new game after playing at night | Clock starts at 09:00 | Passed |
| QA-04 | Sack | Try to buy the sack at 49 items, then at 50 | Refused at 49; bought at 50 for 150 coins | Passed |
| QA-05 | Tools | Buy or equip the Basic Grabber without the sack | Refused; shop shows 🔒 Need sack | Passed |
| QA-06 | Pickup | Send PickupFinish right after PickupBegin (cheat test) | Refused; hold bar resets | Not run |
| QA-07 | Pickup | Finish a pickup from beyond reach + 8 studs | Refused | Not run |
| QA-08 | Pickup | Try to pick up with a full sack | Prompt hidden; "bag full" message | Not run |
| QA-09 | Multi-grab | Pick up with the Basic Grabber next to a cluster | Up to 3 items, nearest within 7 studs, capped by space | Not run |
| QA-10 | Deposit | Deposit a mixed sack at one bin | Only that bin's items go in; coins match the payout formula; toast lists the rest | Not run |
| QA-11 | Deposit | Deposit only wrong items | Nothing moves; message names the right bin and colour | Not run |
| QA-12 | Throw | Face a bin within 30 studs and throw | Item lands inside; credited; Throwing counter +1 | Not run |
| QA-13 | Saves | Throw 5 items onto the floor, exit, continue | The 5 loose items are back in front of the pile | Not run |
| QA-14 | Daily | Claim, then claim again | Second claim refused with time left; streak shown | Not run |
| QA-15 | Progress | Reach 25 items | Banner, sting, +25 coins, Goals row ticked | Not run |

| ID | Area | Test | Expected | Status |
|---|---|---|---|---|
| QA-16 | Robux | Test-buy each pass in Studio | Effect applies at once; shows ✅ OWNED | Not run |
| QA-17 | Robux | Process the same product receipt twice | Granted once only | Not run |
| QA-18 | Saves | Press EXIT, then look at the slot in the menu | Slot shows the right items, coins and time | Not run |
| QA-19 | Saves | Simulate a DataStore failure on join | Player kicked with the "progress is safe" message; save untouched | Not run |
| QA-20 | Security | Fire 30 BuyUpgrade calls in 1 s | About 12–18 processed (12 at once, refilling 6 per second); the rest dropped | Not run |
| QA-21 | Phone | Run with the touch layout | Buttons in a bottom row, coins top right, nothing over the joystick | Passed |
| QA-22 | Phone | Rotate the phone upright in the live game | Game stays landscape | Not run |
| QA-23 | Phone | Play near the pile on a real phone | No lag with all items drawn | Passed |
| QA-24 | PC | Play on PC after the phone layout change | Layout unchanged | Passed |
| QA-25 | Intro | Press SKIP mid-intro | Gate shut, player at landing spot, HUD and controls back | Not run |
| QA-26 | Escape | Dev near-escape, then sort the last 10 | Banner, YOU ESCAPED! screen, WALK OUT leads to the gate | Not run |
| QA-27 | Rebirth | Rebirth after escaping | Progress and pile reset; titles, trails and passes kept; +25% coins | Not run |

# 21. Roadmap, glossary and version history

## 21.1 Roadmap




roadmap · 4 phases, 3 gates, not to scale

Phase 4 lists candidate ideas only; which ones ship depends on what phase 2's data shows.

## 21.2 Glossary

| Term | Meaning |
|---|---|
| Item / piece of trash | One physical object in the pile; each counts 1 toward 100,000 |
| Pile | The 100,000 numbered items in the cage at the start of a sentence |
| Loose item | An item thrown or dropped and not yet sorted; saved with its type |
| Sentence | One run from New Game to escape |
| Sorted | Put in the correct bin; only sorted items count toward escape |
| Grab | Items taken in one pickup |
| Reach | Pickup distance: tool reach + Collection Radius |
| Capacity | Items the player can carry at once |

| Term | Meaning |
|---|---|
| Trash Sack | The one-time 150-coin purchase that lifts capacity from 1 to 3 and unlocks tools |
| Perk | Late-game one-time coin purchase (Auto-Deposit, Auto-Sort Sack) |
| Pass | Permanent Robux purchase |
| Product | Repeatable Robux purchase (coins, boosts, Next Tool, Sack Upgrade, Cleanup Crew) |
| Boost | 15-minute Robux effect that stacks in time |
| Milestone | Reward for total items sorted (18 in all) |
| Challenge | Side goal on one counter (20 in all) |
| Rebirth | Optional restart after escape for +25% coins per rebirth |
| Slot | One of 3 independent saves per account |
| Warden | The unseen PA voice who runs the tutorial |
| Server-authoritative | The server decides every result; the client only asks |

## 21.3 Version history

| Version | Date | Change |
|---|---|---|
| 1.1 | 24 Oct 2026 | Part II added: feature specs, economy model, UI specs, narrative, audio, technical appendix, analytics, accessibility, QA plan, roadmap. Part I corrected: the Warden only speaks in the tutorial; milestones use banners; receipts are stored on the account |
| 1.0 | 23 Oct 2026 | Part I: summary GDD with diagrams, screenshots, item sheet, full config numbers, phone layout and launch checklist |